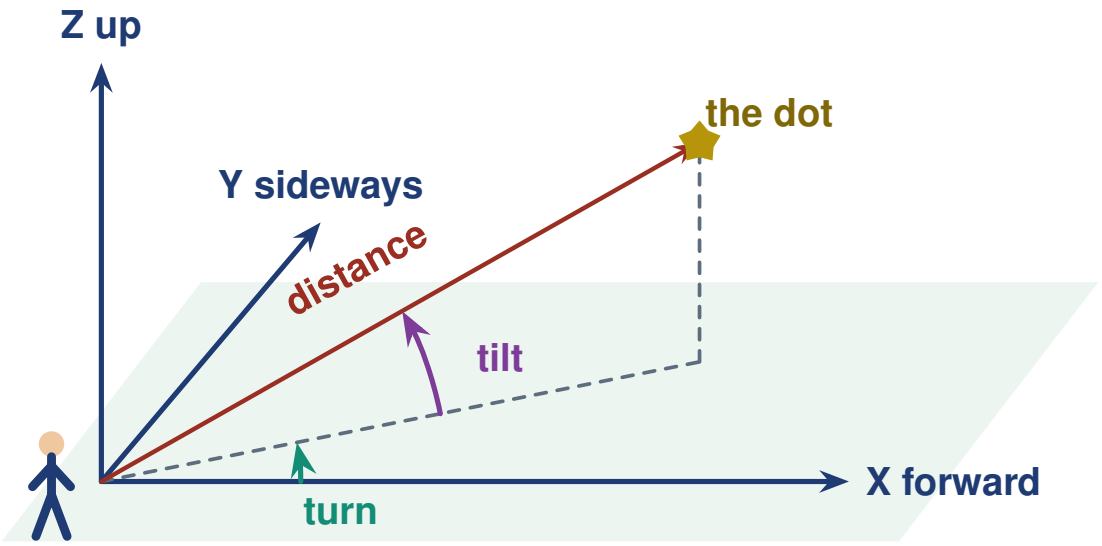
From Angles and Distance to X, Y, Z

BIG IDEA

A turn, a tilt and a distance tell you exactly where a dot is.



THE IDEA

To tell a friend where treasure is hidden, you could say: “Face the big tree. Turn a little to the left. Look a little up. Now walk 25 steps that way.” That is a **turn**, a **tilt** and a **distance**. A LiDAR records the same three things for every dot, then uses triangle math to change them into how far forward, how far sideways and how far up: X, Y and Z.

REMEMBER

- ✓ Every dot starts as a turn, a tilt and a distance.
- ✓ X is forward, Y is sideways and Z is up.
- ✓ Driving faster spreads the lines of dots farther apart on the road.

TRY THIS

Hide a toy in the room. Guide a friend to it using only a turn, a tilt (“look up a little”) and a number of steps.